

Comparison of the Prevalences and Antimicrobial Resistances of Escherichia coli Isolates from Different Retail Meats in the United States, 2002 to 2008

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Escherichia coli isolates were recovered from the National Antimicrobial Resistance Monitoring System retail meat program and examined for antimicrobial susceptibility. Retail meat samples (n = 11,921) from four U.S. states collected during 2002 to 2008, consisting of 2,988 chicken breast, 2,942 ground turkey, 2,991 ground beef, and 3,000 pork chop samples, were analyzed. A total of 8,286 E. coli isolates were recovered. The greatest numbers of samples contaminated with the organism were chicken (83.5%) and turkey (82.0%), followed by beef (68.9%) and pork (44.0%). Resistance was most common to tetracycline (50.3%), followed by streptomycin (34.6%), sulfamethoxazole-sulfisoxazole (31.6%), ampicillin (22.5%), gentamicin (18.6%), kanamycin (8.4%), amoxicillin-clavulanic acid (6.4%), and ceftioxin (5.2%). Less than 5% of the isolates had resistance to trimethoprim, ceftriaxone, ceftiofur, nalidixic acid, chloramphenicol, and ciprofloxacin. All isolates were susceptible to amikacin. Compared to beef and pork isolates, the poultry meat isolates had a greater percentage of resistance to all tested drugs, with the exception of chloramphenicol, to which pork isolates had the most resistance. More than half of the turkey isolates (56%) were resistant to multidrugs (≥ 3 classes) compared to 38.9% of chicken, 17.3% of pork, and 9.3% of beef isolates. The blaCMY gene was present in all ceftriaxone- and ceftiofur-resistant isolates. The cmlA, flo, and catI genes were present in 45%, 43%, and 40% of chloramphenicol-resistant isolates, respectively. Most nalidixic acid-resistant isolates (98.5%) had a gyrA mutation in S83 or D87 or both, whereas only 6.7% had a parC mutation in either S80 or E84. The results showed that E. coli was commonly present in the retail meats, and antimicrobial resistance profiles differed according to the animal origin of the isolates.